MEMORY ANALYSIS REPORT

Memory Forensics — Cobalt Strike Intrusion

Case File: win11_infected.vmem

Field	Detail
Case File	win11_infected.vmem\
Analysis Date	2026-03-15
Analysis	Tao (Juana) Fan
Analyst Tool	Volatility 3 Framework 2.28.0 (FLARE-VM)
Threat Classification	Trojan.CobaltStrike/Zusy — Active Intrusion
Victim Host	Windows 11 (analyst workstation)
C2 Server	192.168.200.128 (port 80, 8080)
License ID (Watermark)	426352781
Severity	PRACTICAL

1. Executive Summary

This report documents a confirmed Cobalt Strike intrusion identified through Windows memory forensics. The victim user (analyst) executed a weaponized version of Advanced IP Scanner (ip_scanner.exe) downloaded from their Downloads directory. The malware established a command-and-control (C2) beacon to 192.168.200.128, performed lateral reconnaissance via IP scanning of the 192.168.200.1-254 subnet, injected a reflective DLL into multiple rundll32.exe processes, and conducted credential theft by injecting code into lsass.exe. No persistent mechanism was found to have been deployed at the time of memory capture, suggesting the image was acquired during an active intrusion.

2. Attack Timeline

Timestamp (UTC)	Event
2023-06-24 00:26:57	User executed legitimate Advanced_IP_Scanner_2.5.4594.1.exe installer (UserAssist record)
2023-06-24 00:27:33	Advanced IP Scanner wrote UI configuration to registry (famatech\advanced_ip_scanner)
2023-06-24 00:51:18	System scheduled tasks last executed (baseline activity)

2023-06-24 00:58:49	Malicious ip_scanner.exe (PID 1652) launched by explorer.exe
2023-06-24 01:00:50	rundll32.exe (PID 268) spawned by ip_scanner.exe (PID 1652) — no cmdline args, MZARUH beacon injected, C2 on port 80
2023-06-24 01:03:51	rundll32.exe (PID 4704) spawned by rundll32.exe (PID 268) — MZARUH beacon injected, C2 on port 8080
2023-06-24 01:09:36	Second instance of ip_scanner.exe (PID 10112) launched — MZARUH beacon present, no C2 connection observed
2023-06-24 01:18:24	WsSwapAssessmentTask last run (system task — coincidental timing)
2023-06-24 01:31:45	Last registry write by Advanced IP Scanner — scan range 192.168.200.1-254 recorded
2023-06-24 01:31:47	AppCompatFlags records both ip_scanner.exe and legitimate installer as executed
Time of capture	Memory image acquired — no persistence mechanism deployed yet

3. Process Tree Analysis

The following process hierarchy was identified as malicious:

```
PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.pstree --pid 10112
Volatility 3 Framework 2.28.0
Formatting...0.00          PDB scanning finished
      |   PID | PPID |  ImageFileName |     Offset(V) | Threads | Handles | SessionId | Wow64 |                    CreateTime |                    ExitTime |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
 |   PID | PPID |  ImageFileName |     Offset(V) | Threads | Handles | SessionId | Wow64 |                    CreateTime |                    ExitTime |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
 |   PID | PPID |  ImageFileName |     Offset(V) | Threads | Handles | SessionId | Wow64 |                    CreateTime |                    ExitTime |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
 |   PID | PPID |  ImageFileName |     Offset(V) | Threads | Handles | SessionId | Wow64 |                    CreateTime |                    ExitTime |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
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Key observations:

- Both rundll32.exe instances were launched with empty command-line arguments, which is anomalous as rundll32.exe requires a DLL argument to perform any legitimate function.
- rundll32.exe spawning rundll32.exe is a strong indicator of process injection or parent process spoofing.
- All four malicious processes contained MZARUH magic bytes (4D 5A 41 52 55 48), a known Cobalt Strike reflective DLL injection signature.

```
PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.malware.malfind --pid 10112
Volatility 3 Framework 2.28.0
Formatting...0.00          PDB scanning finished
      PID |          Process | Start VPN |  End VPN |  Tag |            Protection | CommitCharge | PrivateMemory | File output |     Notes |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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* | 10112 | ip_scanner.exe |  0x650000 | 0x69dfff | VadS | PAGE_EXECUTE_READWRITE |           78 |             1 |   Disabled | MZ header |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.malware.malfind --pid 268
Volatility 3 Framework 2.28.0
Formatting...0.00               PDB scanning finished
  | PID |      Process |     Start VPN |       End VPN |  Tag |              Protection | CommitCharge | PrivateMemory | File output |      Notes |                                                                                                                                                Hexdump |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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* | 268 | rundll32.exe | 0x18c124f0000 | 0x18c12530fff | VadS | PAGE_EXECUTE_READWRITE |           65 |             1 |    Disabled | MZ header |
| |     |              |               |               |      |                        |              |               |             |           |
| |     |              |               |               |      |                        |              |               |             |           | 4d 5a 41 52 55 48 89 e5 48 81 ec 20 00 00 00 48 MZARUH..H.. ...H |                 0x18c124f0000:  pop     r10
| |     |              |               |               |      |                        |              |               |             |           | 8d 1d ea ff ff ff 48 89 df 48 81 c3 f4 5f 01 00 ......H..H..._.. |                 0x18c124f0002:  push    r10
| |     |              |               |               |      |                        |              |               |             |           | ff d3 41 b8 f0 b5 a2 56 68 04 00 00 00 5a 48 89 ..A....Vh....ZH. |                 0x18c124f0004:  push    rbp
| |     |              |               |               |      |                        |              |               |             |           | f9 ff d0 00 00 00 00 00 00 00 00 00 f8 00 00 00 ................ |            0x18c124f0005:  mov     rbp, rsp
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |          0x18c124f0008:  sub     rsp, 0x20
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |   0x18c124f000f:  lea     rbx, [rip - 0x16]
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |            0x18c124f0016:  mov     rdi, rbx
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |        0x18c124f0019:  add     rbx, 0x15ff4
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                0x18c124f0020:  call    rbx
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |    0x18c124f0022:  mov     r8d, 0x56a2b5f0
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                  0x18c124f0028:  push    4
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                0x18c124f002d:  pop     rdx
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |           0x18c124f002e:  mov     rcx, rdi
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                0x18c124f0031:  call    rax
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c124f0033:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c124f0035:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c124f0037:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c124f0039:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |            0x18c124f003b:  add     al, bh
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c124f003d:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |
* | 268 | rundll32.exe | 0x18c127b0000 | 0x18c127fdfff | VadS | PAGE_EXECUTE_READWRITE |           78 |             1 |    Disabled | MZ header |
| |     |              |               |               |      |                        |              |               |             |           |
| |     |              |               |               |      |                        |              |               |             |           | 4d 5a 41 52 55 48 89 e5 48 81 ec 20 00 00 00 48 MZARUH..H.. ...H |                 0x18c127b0000:  pop     r10
| |     |              |               |               |      |                        |              |               |             |           | 8d 1d ea ff ff ff 48 89 df 48 81 c3 f4 5f 01 00 ......H..H..._.. |                 0x18c127b0002:  push    r10
| |     |              |               |               |      |                        |              |               |             |           | ff d3 41 b8 f0 b5 a2 56 68 04 00 00 00 5a 48 89 ..A....Vh....ZH. |                 0x18c127b0004:  push    rbp
| |     |              |               |               |      |                        |              |               |             |           | f9 ff d0 00 00 00 00 00 00 00 00 00 f8 00 00 00 ................ |            0x18c127b0005:  mov     rbp, rsp
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |          0x18c127b0008:  sub     rsp, 0x20
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |   0x18c127b000f:  lea     rbx, [rip - 0x16]
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |            0x18c127b0016:  mov     rdi, rbx
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |        0x18c127b0019:  add     rbx, 0x15ff4
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                0x18c127b0020:  call    rbx
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |    0x18c127b0022:  mov     r8d, 0x56a2b5f0
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                  0x18c127b0028:  push    4
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                0x18c127b002d:  pop     rdx
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |           0x18c127b002e:  mov     rcx, rdi
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |                0x18c127b0031:  call    rax
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c127b0033:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c127b0035:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c127b0037:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c127b0039:  add     byte ptr [rax], al
| |     |              |               |               |      |                        |              |               |             |           |                                                                  |            0x18c127b003b:  add     al, bh
| |     |              |               |               |      |                        |              |               |             |           |                                                                  | 0x18c127b003d:  add     byte ptr [rax], al
FLARE-VM 03/15/2026 03:12:07
```

```
PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.malware.malfind --pid 4704
Volatility 3 Framework 2.28.0
Formatting...0.00               PDB scanning finished
  | PID |      Process |     Start VPN |       End VPN |  Tag |              Protection | CommitCharge | PrivateMemory | File output |      Notes |                                                                                                                                                Hexdump |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
* | 4704 | rundll32.exe | 0x2876e3b0000 | 0x2876e3f0fff | VadS | PAGE_EXECUTE_READWRITE |           65 |             1 |    Disabled | MZ header |
| |      |              |               |               |      |                        |              |               |             |           | 4d 5a 41 52 55 48 89 e5 48 81 ec 20 00 00 00 48 MZARUH..H.. ...H |                 0x2876e3b0000:  pop     r10
| |      |              |               |               |      |                        |              |               |             |           | 8d 1d ea ff ff ff 48 89 df 48 81 c3 f4 5f 01 00 ......H..H..._.. |                 0x2876e3b0002:  push    r10
| |      |              |               |               |      |                        |              |               |             |           | ff d3 41 b8 f0 b5 a2 56 68 04 00 00 00 5a 48 89 ..A....Vh....ZH. |                 0x2876e3b0004:  push    rbp
| |      |              |               |               |      |                        |              |               |             |           | f9 ff d0 00 00 00 00 00 00 00 00 00 f8 00 00 00 ................ |            0x2876e3b0005:  mov     rbp, rsp
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |          0x2876e3b0008:  sub     rsp, 0x20
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |   0x2876e3b000f:  lea     rbx, [rip - 0x16]
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |            0x2876e3b0016:  mov     rdi, rbx
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |        0x2876e3b0019:  add     rbx, 0x15ff4
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                0x2876e3b0020:  call    rbx
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |    0x2876e3b0022:  mov     r8d, 0x56a2b5f0
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                  0x2876e3b0028:  push    4
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                0x2876e3b002d:  pop     rdx
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |           0x2876e3b002e:  mov     rcx, rdi
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                0x2876e3b0031:  call    rax
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e3b0033:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e3b0035:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e3b0037:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e3b0039:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |            0x2876e3b003b:  add     al, bh
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e3b003d:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |
* | 4704 | rundll32.exe | 0x2876e570000 | 0x2876e5bdfff | VadS | PAGE_EXECUTE_READWRITE |           78 |             1 |    Disabled | MZ header |
| |      |              |               |               |      |                        |              |               |             |           |
| |      |              |               |               |      |                        |              |               |             |           | 4d 5a 41 52 55 48 89 e5 48 81 ec 20 00 00 00 48 MZARUH..H.. ...H |                 0x2876e570000:  pop     r10
| |      |              |               |               |      |                        |              |               |             |           | 8d 1d ea ff ff ff 48 89 df 48 81 c3 f4 5f 01 00 ......H..H..._.. |                 0x2876e570002:  push    r10
| |      |              |               |               |      |                        |              |               |             |           | ff d3 41 b8 f0 b5 a2 56 68 04 00 00 00 5a 48 89 ..A....Vh....ZH. |                 0x2876e570004:  push    rbp
| |      |              |               |               |      |                        |              |               |             |           | f9 ff d0 00 00 00 00 00 00 00 00 00 f8 00 00 00 ................ |            0x2876e570005:  mov     rbp, rsp
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |          0x2876e570008:  sub     rsp, 0x20
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |   0x2876e57000f:  lea     rbx, [rip - 0x16]
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |            0x2876e570016:  mov     rdi, rbx
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |        0x2876e570019:  add     rbx, 0x15ff4
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                0x2876e570020:  call    rbx
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |    0x2876e570022:  mov     r8d, 0x56a2b5f0
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                  0x2876e570028:  push    4
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                0x2876e57002d:  pop     rdx
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |           0x2876e57002e:  mov     rcx, rdi
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |                0x2876e570031:  call    rax
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e570033:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e570035:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e570037:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e570039:  add     byte ptr [rax], al
| |      |              |               |               |      |                        |              |               |             |           |                                                                  |            0x2876e57003b:  add     al, bh
| |      |              |               |               |      |                        |              |               |             |           |                                                                  | 0x2876e57003d:  add     byte ptr [rax], al
FLARE-VM 03/15/2026 03:13:06
```

4. Cobalt Strike Beacon Configuration

PID	Process	Port	URI	Sleep (ms)	License ID
1652	ip_scanner.exe	8080	/push	60000	426352781
268	rundll32.exe	80	/ptj	60000	426352781
4704	rundll32.exe	8080	/pixel	60000	426352781
10112	ip_scanner.exe	8080	/push	60000	426352781

Notable configuration details:

- All four beacons share License ID 426352781, confirming they originate from the same Cobalt Strike Team Server.
- Multiple URI paths (/push, /ptj, /pixel) indicate a Malleable C2 profile designed to mimic legitimate HTTP traffic.
- Jitter is set to 0, creating a perfectly regular 60-second heartbeat — an operational security weakness that simplifies network-based detection.
- Install paths configured as %windir%\syswow64\rundll32.exe and %windir%\sysnative\rundll32.exe indicate planned persistence via rundll32, though not yet executed at time of capture.

```
PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.cobaltstrike.CobaltStrike
Volatility 3 Framework 2.28.0
Formatting...0.00               PDB scanning finished
  |   PID |        Process | Port | Sleep | Jitter |                 Server |   POST_PATH |              x86 Install_Path |              x64 Install_Path | Pipe | License ID
* |  1652 | ip_scanner.exe | 8080 | 60000 |      0 |  192.168.200.128,/push | /submit.php | %windir%\syswow64\rundll32.exe | %windir%\sysnative\rundll32.exe |      |  426352781
* |   268 |   rundll32.exe |   80 | 60000 |      0 |   192.168.200.128,/ptj | /submit.php | %windir%\syswow64\rundll32.exe | %windir%\sysnative\rundll32.exe |      |  426352781
* |  4704 |   rundll32.exe | 8080 | 60000 |      0 | 192.168.200.128,/pixel | /submit.php | %windir%\syswow64\rundll32.exe | %windir%\sysnative\rundll32.exe |      |  426352781
* | 10112 | ip_scanner.exe | 8080 | 60000 |      0 |  192.168.200.128,/push | /submit.php | %windir%\syswow64\rundll32.exe | %windir%\sysnative\rundll32.exe |      |  426352781
FLARE-VM 03/15/2026 04:02:00
```

5. Credential Theft — lsass.exe Injection

Malfind analysis of lsass.exe (PID 672) identified two anomalous PAGE_EXECUTE_READWRITE memory regions not associated with any loaded module:

Address	Observation
0x1c9c14c0000	Contains plaintext string 'samsrv.d' — references samsrv.dll, which handles SAM database access. Consistent with credential harvesting targeting the Security Accounts Manager.
0x1c9c15f0000	Contains injected x64 function code (full register prologue: push rbp/rbx/rsi/rdi/r12-r15). Appears to be a complete injected function, likely responsible for credential extraction logic.

```
PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.malware.malfind --pid 672
C:\Tools\volatility3-develop\volatility3\framework\deprecation.py:105: FutureWarning: This plugin (PluginRequirement) has been renamed and will be removed in the first release after 2026-06-01. PluginRequirement is to be deprecated. Use VersionRequirement instead.
  warnings.warn(
Volatility 3 Framework 2.28.0
Formatting...0.00               PDB scanning finished
  | PID |   Process |     Start VPN |       End VPN |  Tag |            Protection | CommitCharge | PrivateMemory | File output | Notes |                                                                                Hexdump |                                                                                Disasm
* | 672 | lsass.exe | 0x1c9c14c0000 | 0x1c9c14c0fff | VadS | PAGE_EXECUTE_READWRITE |            1 |             1 |    Disabled |   N/A |
  |     |           |               |               |      |                        |              |               |             |       | 70 0c c7 86 fc 7f 00 00 50 b6 c6 86 fc 7f 00 00 p.......P....... |                                0x1c9c14c0000:  jo      0x1c9c14c000e
  |     |           |               |               |      |                        |              |               |             |       | 50 cf c6 86 fc 7f 00 00 80 51 c7 86 fc 7f 00 00 P........Q...... | 0x1c9c14c0002:  mov     dword ptr [rsi + 0x7ffc], 0x86c6b650
  |     |           |               |               |      |                        |              |               |             |       | 00 52 c7 86 fc 7f 00 00 60 50 c7 86 fc 7f 00 00 .R......`P...... |                                0x1c9c14c000c:  cld
  |     |           |               |               |      |                        |              |               |             |       | 50 52 c7 86 fc 7f 00 00 73 61 6d 73 72 76 2e 64 PR......samsrv.d |                                0x1c9c14c000d:  jg      0x1c9c14c000f
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                         0x1c9c14c000f:  add     byte ptr [rax - 0x31], dl
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c14c0012:  mov     byte ptr [rsi + 0x7ffc], 0x80
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                    0x1c9c14c0019:  push    rcx
  |     |           |               |               |      |                        |              |               |             |       |                                                                  | 0x1c9c14c001a:  mov     dword ptr [rsi + 0x7ffc], 0x86c75200
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                    0x1c9c14c0024:  cld
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c14c0025:  jg      0x1c9c14c0027
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                         0x1c9c14c0027:  add     byte ptr [rax + 0x50], ah
  |     |           |               |               |      |                        |              |               |             |       |                                                                  | 0x1c9c14c002a:  mov     dword ptr [rsi + 0x7ffc], 0x86c75250
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                    0x1c9c14c0034:  cld
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c14c0035:  jg      0x1c9c14c0037
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                         0x1c9c14c0037:  add     byte ptr [rbx + 0x61], dh
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c14c003a:  insd    dword ptr [rdi], dx
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c14c003b:  jae     0x1c9c14c00af
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c14c003d:  jbe     0x1c9c14c006d
* | 672 | lsass.exe | 0x1c9c15f0000 | 0x1c9c15f0fff | VadS | PAGE_EXECUTE_READWRITE |            1 |             1 |    Disabled |   N/A |
  |     |           |               |               |      |                        |              |               |             |       | 40 55 53 56 57 41 54 41 55 41 56 41 57 48 8d 6c @USVWATAUAVAWH.l |                                0x1c9c15f0000:  push    rbp
  |     |           |               |               |      |                        |              |               |             |       | 24 d8 48 81 ec 28 01 00 00 45 33 f6 48 8b d9 48 $.H..(...E3.H..H |                                0x1c9c15f0002:  push    rbx
  |     |           |               |               |      |                        |              |               |             |       | 83 c1 38 4c 89 74 24 58 4c 89 75 d8 4c 89 74 24 ..8L.t$XL.u.L.t$ |                                0x1c9c15f0003:  push    rsi
  |     |           |               |               |      |                        |              |               |             |       | 40 4c 89 74 24 38 4c 89 74 24 30 44 89 75 70 4c @L.t$8L.t$0D.upL |                                0x1c9c15f0004:  push    rdi
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f0005:  push    r12
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f0007:  push    r13
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f0009:  push    r14
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f000b:  push    r15
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c15f000d:  lea     rbp, [rsp - 0x20]
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f0012:  sub     rsp, 0x128
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f0019:  xor     r14d, r14d
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f001c:  mov     rbx, rcx
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                                0x1c9c15f001f:  add     rcx, 0x38
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c15f0023:  mov     qword ptr [rsp + 0x58], r14
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c15f0028:  mov     qword ptr [rsp - 0x20], r14
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c15f002c:  mov     qword ptr [rsp + 0x40], r14
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c15f0031:  mov     qword ptr [rsp + 0x38], r14
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                    0x1c9c15f0036:  mov     qword ptr [rsp + 0x30], r14
  |     |           |               |               |      |                        |              |               |             |       |                                                                  |                0x1c9c15f003b:  mov     dword ptr [rbp + 0x70], r14d
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```

lsass.exe showed no outbound network connections, which is expected: credential data is exfiltrated by the Cobalt Strike beacon (rundll32.exe) via the established C2 channel, not directly by lsass.

6. Network Indicators

Process (PID)	Remote IP	Port	Notes
ip_scanner.exe (1652)	192.168.200.128	8080	C2 beacon — connection closed at capture time
rundll32.exe (268)	192.168.200.128	80	C2 beacon — connection closed at capture time
rundll32.exe (4704)	192.168.200.128	8080	C2 beacon — connection closed at capture time
ip_scanner.exe (10112)	—	—	No C2 connection observed
lsass.exe (672)	—	—	Listening only — no outbound connections (expected)

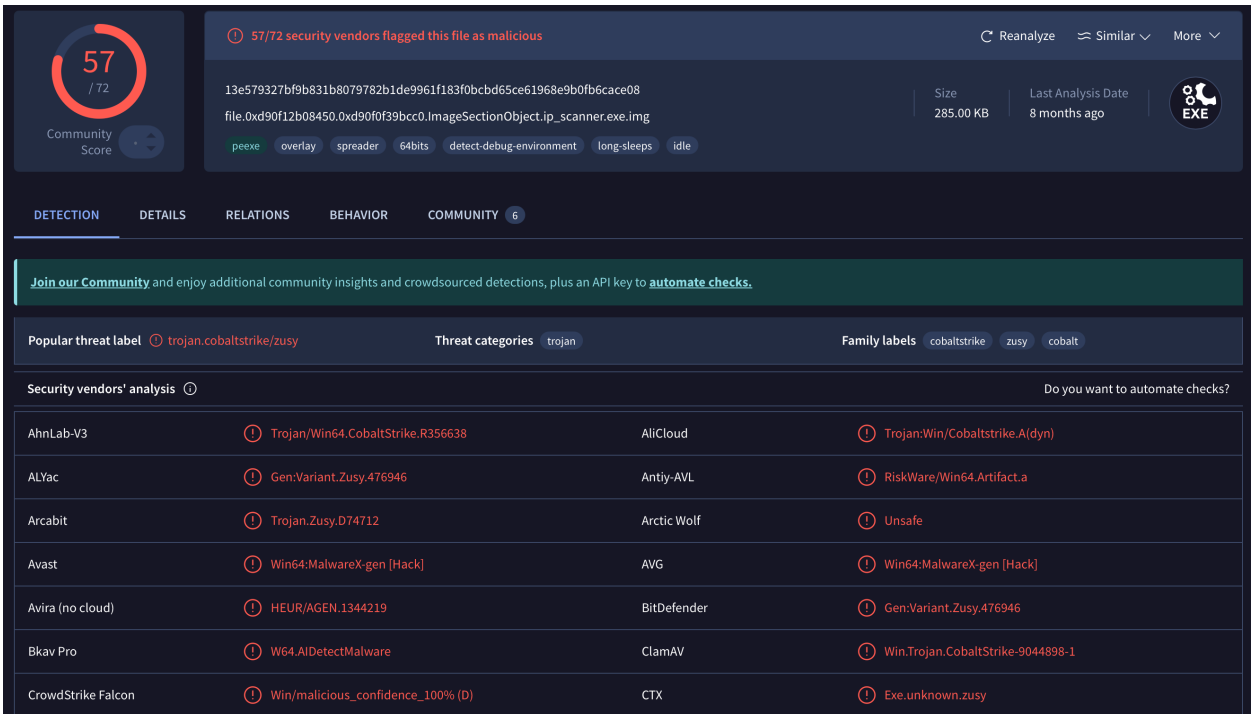
```
PS C:\Users\FlareVM\Desktop\Mem\win11_infected > vol3.py -r pretty -f .\win11_infected.vmem windows.netscan > netscan.txt
|   |         Offset | Proto |              LocalAddr | LocalPort |     ForeignAddr | ForeignPort |       State |   PID |          Owner |                       Created
* | 0xd90f0beb1010 | TCPv4 |          192.168.200.133 |     52223 | 192.168.200.128 |          80 |      CLOSED |   268 |   rundll32.exe | 2023-06-24 01:23:53.000000 UTC
* | 0xd90f0bfe7010 | TCPv4 |          192.168.200.133 |     52229 | 192.168.200.128 |          80 |      CLOSED |   268 |   rundll32.exe | 2023-06-24 01:24:53.000000 UTC
* | 0xd90f0c356010 | TCPv4 |          192.168.200.133 |     52294 | 192.168.200.128 |          80 |  CLOSE_WAIT |   268 |   rundll32.exe | 2023-06-24 01:32:53.000000 UTC
* | 0xd90f0f903010 | TCPv4 |          192.168.200.133 |     52256 | 192.168.200.128 |          80 |      CLOSED |   268 |   rundll32.exe | 2023-06-24 01:29:53.000000 UTC
* | 0xd90f0f914010 | TCPv4 |          192.168.200.133 |     52218 | 192.168.200.128 |          80 |      CLOSED |   268 |   rundll32.exe | 2023-06-24 01:22:53.000000 UTC
* | 0xd90f0c353010 | TCPv4 |          192.168.200.133 |     52292 | 192.168.200.128 |        8080 |      CLOSED |  4704 |   rundll32.exe | 2023-06-24 01:32:52.000000 UTC
* | 0xd90f176e9010 | TCPv4 |          192.168.200.133 |     52247 | 192.168.200.128 |        8080 |      CLOSED | 10112 | ip_scanner.exe | 2023-06-24 01:28:45.000000 UTC
* | 0xd90f0b921010 | TCPv4 |          192.168.200.133 |     52232 | 192.168.200.128 |        8080 |      CLOSED | 10112 | ip_scanner.exe | 2023-06-24 01:25:43.000000 UTC
* | 0xd90f197f3010 | TCPv4 |          192.168.200.133 |     52258 | 192.168.200.128 |        8080 |      CLOSED | 10112 | ip_scanner.exe | 2023-06-24 01:30:47.000000 UTC
```

7. Reconnaissance Activity

Registry analysis of the famatech\advanced_ip_scanner hive revealed that the analyst account ran both the legitimate Advanced IP Scanner and the trojanized ip_scanner.exe. The key value LastRangeUsed recorded a scan of 192.168.200.1-254, which directly corresponds to the subnet containing the C2 server (192.168.200.128). This indicates the attacker used the IP scanner functionality to perform internal network reconnaissance prior to or concurrent with beacon execution.

8. Malware Identification

Field	Detail
File	ip_scanner.exe (ImageSection dump)
SHA-256	13e579327bf9b831b8079782b1de9961f183f0bcbd65ce61968e9b0fb6cace08
File Size	285.00 KB
VirusTotal Detection	57/72 vendors
Threat Label	trojan.cobaltstrike/zusy
Family Labels	cobaltstrike, zusy, cobalt
Behavioral Tags	peexe, overlay, spreader, 64bits, detect-debug-environment, long-sleeps, idle
Notable Tags	spreader — indicates lateral movement capability consistent with IP scanner disguise



9. Persistence Assessment

No active persistence mechanism was identified at the time of memory capture. The following locations were examined and found clean:

- Registry Run Keys (HKCU and HKLM)
- Scheduled Tasks
- Windows Services (svcscan)
- WMI Subscriptions

Assessment: Given that the analyst account held local administrator privileges, the absence of persistence suggests the memory image was captured during an active session before the attacker completed their objectives. The Cobalt Strike beacon configuration includes pre-set install paths, indicating persistence deployment was planned but not yet executed.

10. Indicators of Compromise (IOCs)

Type	Value	Context
SHA-256	13e579327bf9b831b8079782b1de9961f183f0bcbd65ce61968e9b0fb6cace08	ip_scanner.exe image section
IP Address	192.168.200.128	Cobalt Strike C2 server
TCP	80, 8080	C2

Port		communication ports
URI URI	/push, /ptj, /pixel /submit.php	Malleable C2 beacon URIs C2 POST path
CS License ID	426352781	Team Server watermark — use for correlation
Memory Sig	4D 5A 41 52 55 48 (MZARUH)	Cobalt Strike reflective DLL magic bytes
File Path	C:\Users\analyst\Downloads\ip_scanner.exe	Malicious executable location
Process Process Chain	rundll32.exe with empty cmdline ip_scanner.exe → rundll32.exe → rundll32.exe	Behavioral IOC Behavioral IOC — process injection chain

11. Recommended Response Actions

- Isolate the affected host immediately to prevent further C2 communication or lateral movement.
- Reset all credentials associated with the analyst account — assume all passwords accessible from this host are compromised given the lsass injection.
- Investigate 192.168.200.128 — identify the C2 server owner, determine if it is internal infrastructure that has been compromised.
- Search for CS License ID 426352781 across threat intelligence platforms to identify related campaigns.
- Hunt for MZARUH memory signature and empty-cmdline rundll32 processes across the environment.
- Block C2 IOCs (IP, ports, URIs) at the perimeter firewall and proxy.
- Conduct disk forensics on the affected host to recover deleted files and identify the full infection chain, including the original download source of ip_scanner.exe.
- Review network logs for beacon traffic patterns (60-second intervals, zero jitter) to identify other potentially compromised hosts.